

Each question is worth 5 points for a total of 40 points. The Extra Credit problem can get you up to an additional 5 points.

**Problem 1**

Find the volume of the solid generated by revolving the region bound by the curves  $y = x^2$ ,  $y = 4$ , and  $x = 0$  about the  $y$ -axis.

**Problem 2**

Find the volume of the solid generated by revolving the region bound by the curves  $y = \frac{1}{2}x$  and  $y = \sqrt{x}$  about the  $x$ -axis.

**Problem 3**

Find the lengths of the curve  $y = \frac{x^3}{12} + \frac{1}{x}$  with  $1 \leq x \leq 4$

**Problem 4**

Find the AREA (not volume) of the surface generated by revolving the curve  $y = 2\sqrt{x}$ , with  $1 \leq x \leq 2$ , about the  $x$ -axis.

**Problem 5**

Let  $f(x) = \frac{x+3}{x-2}$ . Find  $f^{-1}(x)$ , the domains of  $f(x)$  and  $f^{-1}(x)$ , and the ranges of  $f(x)$  and  $f^{-1}(x)$ . Then find  $f(f^{-1}(x))$  and  $f^{-1}(f(x))$ .

**Problem 6**

Solve

$$\int \frac{t}{t^2 + 16} dt.$$

**Problem 7**

Solve

$$\int_{\ln(8)}^{\ln(27)} e^{x/3} dx.$$

**Problem 8**

Solve the differential equation

$$2\sqrt{x}\frac{dy}{dx} = \frac{1}{\sqrt{y}}.$$

**Extra Credit:**

Suppose that the bacteria in a colony can grow unchecked, by the law of exponential change. The colony starts with 1 bacterium and doubles every half-hour. How many bacteria will the colony contain at the end of 24 hours?